

ANGLO-CHINESE JUNIOR COLLEGE
MATHEMATICS DEPARTMENT

MATHEMATICS
Higher 1
Paper 1

8863 / 01

18 August 2009

JC 2 PRELIMINARY EXAMINATION

Time allowed: **3 hours**

Additional Materials: List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Index number, Form Class, graphic and/or scientific calculator model/s on the cover page.

Write your Index number and full name on all the work you hand in.

Write in dark blue or black pen on your answer scripts.

You may use a soft pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in the question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This document consists of 7 printed pages.



Anglo-Chinese Junior College

[Turn Over]

ANGLO-CHINESE JUNIOR COLLEGE
MATHEMATICS DEPARTMENT
JC2 Preliminary Examination 2009

MATHEMATICS 8863
Higher 1
Paper 1

/ 100

Index No:

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Form Class: _____

Name: _____

Calculator model: _____

Arrange your answers in the same numerical order.

Place this cover sheet on top of them and tie them together with the string provided.

Question no.	Marks
1	
2	
3	
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9	
10	
11	
12	
13	

Section A : Pure Mathematics [40 marks]

1 (a) Find $\int \frac{4\sqrt{x}-10}{x} dx$. [2]

(b) Solve the equation $5^{2x} = e^{kx}$, where k is a constant to be determined exactly.

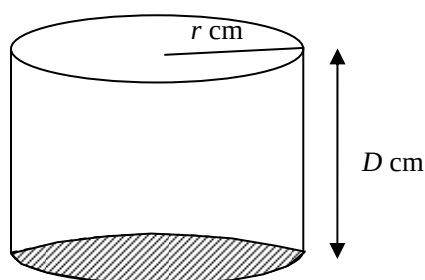
Hence find $\int 5^{2x} dx$. [3]

2 This year, the sum of thrice of Cheryl's age and five times that of Diana's is 110, while the sum of squares of their ages is 386. If Cheryl is younger than Diana, find Cheryl's age in five years' time. [5]

3 Sketch the graphs of $y = \frac{3x+1}{x-1}$ and $y = |x^2 - 3|$ on the same diagram, showing clearly any asymptotes, axial intercepts and stationary points. [5]
Hence solve the inequality

$$\frac{3x+1}{|x^2-3|(x-1)} > 1. \quad [2]$$

4 The diagram shows a cylinder with radius r cm and depth D cm. The cylinder is sealed at the bottom and opened at the top. It has a fixed internal surface area A cm².



Show that the volume of the cup, V cm³, is given by

$$V = \frac{r(A - \pi r^2)}{2}. \quad [2]$$

Show that $D = r$ when the volume is at its maximum. [4]

[Turn Over]

5 The functions f and g are defined as follows:

$$f : x \mapsto \ln x, \quad x \in \mathbb{R}^+$$

$$g : x \mapsto \frac{1}{1+x^2}, \quad x \in \mathbb{R}$$

(i) Explain why g^{-1} does not exist. [1]

(ii) If the domain of g is restricted to the subset of $\mathbb{R}$ for which $x \geq A$, find the least value of A for which g^{-1} exist. [1]

Using this restricted domain of g ,

(iii) find g^{-1} , giving its rule and domain; [3]

(iv) determine if fg exists, justifying your answer. If fg exists, give its rule and domain. [3]

6 (i) Express $\ln\left(\frac{e^{2x}}{\sqrt{1+e^{2x}}}\right)$ in the form $ax + b\ln(1+e^{2x})$, where a and b are constants to be determined. [2]

(ii) The equation of a curve C is given by $y = \ln\left(\frac{e^{2x}}{\sqrt{1+e^{2x}}}\right)$. Using differentiation, show that the equation of the normal to C at the point $x = 0$ is given by

$$y - \ln \frac{1}{\sqrt{2}} = -\frac{2}{3}x. \quad [4]$$

(iii) This normal to C at $x = 0$ cuts the x -axis at point A , and the curve cuts the x -axis at point B . By finding the coordinates of A and B , evaluate the area of the region enclosed by this normal, the x -axis and the curve C . [5]

Section B : Probability and Statistics [60 marks]

- 7 In a school of 1500 students, there are 825 year one students and 675 year two students. An administrative officer from the school wishes to conduct a survey on the preference for the type of food sold to the students. A sample of 50 students is needed for the survey. On one particular day, she interviewed a sample of students from those buying lunch by selecting at random one of the first 30 students buying lunch, then selecting every 30th student after the first student had been chosen.
- (i) Identify the above sampling method. [1]
 - (ii) Give a reason why the officer's method of sampling may be unsatisfactory. [1]
 - (iii) Give a reason why stratified sampling might be preferable in this context and describe how it can be done. [2]
- 8 In a Current Affair's Quiz for 200 JC2 students, there are 20 questions. Each question is followed by five possible answers, of which only one is correct. The students have to decide for each question, which one of the five answers is correct. Assuming that all the students answered the questions randomly,
- (i) find the probability that a randomly selected student has at least 15 incorrect answers; [2]
 - (ii) find, using a suitable approximation, the probability that the total number of correct answers given by all JC2 students is at most 820. [3]
- 9 In a manufacturing process, circular cylinders are being produced as components of a certain product. For a cylinder produced to be acceptable, its length must be between 8.45 cm and 8.65 cm, and its diameter must be between 1.55 cm and 1.60 cm. The cylinders have lengths which are normally distributed with a mean of 8.54 cm and standard deviation 0.05 cm, while independently, the diameters are normally distributed with a mean 1.57 cm and standard deviation 0.01 cm.
- (i) Find the probability that a randomly selected cylinder has unacceptable length. [2]
 - (ii) Find the probability that a randomly selected cylinder has unacceptable diameter. [2]
 - (iii) Hence, show that out of 80 randomly selected cylinders, 5.82 of them are expected to be unacceptable. [3]
 - (iv) Hence, by using a suitable approximation, find the probability that there are at least 10 unacceptable cylinders in a random sample of 80 cylinders. [4]

[Turn Over]

- 10** A manufacturer developed a new mobile phone model, which he claims has a superior mean 'standby-time'. It is known that mobile phones that are currently available have mean 'standby-time' of 15 hours. Given that a random sample of 50 new mobile phones yields the following results for the 'standby-time', x hours, summarised by

$$\sum (x - 8) = 416 \quad \text{and} \quad \sum (x - 8)^2 = 4280,$$

- (i) Find the unbiased estimates for the population mean and variance. **[3]**
- (ii) Test the manufacturer's claim at the 5% level of significance. **[4]**
- (iii) A consumer protection group conducts its own test, using the alternative hypothesis that the mean 'standby-time' of the manufacturer's mobile phone is less than 15 hours, with a standard deviation of 0.5. A sample size of 50 new mobile phones is used. Find the smallest level of significance of the test if the null hypothesis is rejected whenever the sample mean is found to be less than 14.87 hours. **[3]**
- 11** Two machines, A and B , which operate independently, are used to produce identical items. For each item produced by machine A , the probability of it being rejected is 0.02, while for each item produced by machine B , the corresponding probability is 0.05.
- (i) Four items are randomly chosen from those produced by machine A and another two items are randomly chosen from those produced by machine B . Calculate the probability that exactly one of the six chosen items is rejected. **[3]**
- Of the total output of items produced in a day, 55% are produced by machine A and 45% are produced by machine B .
- (ii) If one of the items is chosen at random from a day's output, show that the probability of it not being rejected is 0.9665. **[2]**
- (iii) Three items are chosen at random from a day's output. Find the probability that one is produced by machine A and the other two are produced by machine B , given that all three items are not rejected. **[3]**
- 12** A party was attended by the following thirteen people: a pair of twin sisters and their boyfriends, all from Australia; three female teachers from China; a family consists of a man, his wife, his parents and their two daughters, all from Malaysia. Two people from the group are randomly chosen to play a game. Find the probability that
- (i) the two people were all of the same gender; **[2]**
- (ii) the two people were all of different nationalities; **[3]**
- (iii) the two people were all of different nationalities, given that they were of the same gender. **[5]**

- 13 A study was conducted to determine if there is any correlation between the number of hours office workers spent playing computer games in a week and the number of mistakes they made in their work. To investigate this trend, 8 office workers from various departments were selected. The number of hours, x , they spent per week playing computer games and the number of mistakes, y , they made over a week are tabulated.

x	3.5	4	5	3.8	6	9.4	7.1	8.1
y	6	15	40	49	65	135	70	85

- (a) Calculate the linear product moment correlation coefficient between x and y , and comment on the value found. [2]
- (b) Sketch the scatter diagram for the data. One of the pairs of data appears to be recorded wrongly. Indicate the corresponding point on your diagram by labelling it P , and explain why the scatter diagram for the remaining points may be consistent with a linear model. [4]
- (c) In place of the wrong data point, a **new** pair of data was added such that the equation of the least squares regression line of x on y is found to be $x = 3.46 + 0.0468y$. Given that $\bar{x} = 6.08$, and the new linear product moment correlation coefficient between x and y is 0.973, find the equation of the least squares regression line of y on x . [3]
- (d) Use an appropriate regression line to estimate the following:
- (i) the number of mistakes per week by a worker who plays 6.5 hours of computer games each week; [1]
 - (ii) the number of hours of playing computer games per week if the worker makes 150 mistake that week. [1]
- (e) Comment on the reliability of the estimates in part (d). [1]

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